

Comparing Two, Paired, and Many Means

IMS Chapters 20, 21 & 22

Comparing two independent means

Extending Chapter 19

Chapter 19 handled one mean. Now: two **independent** groups — different, unrelated individuals in each.

Point estimate: $\bar{x}_1 - \bar{x}_2$

Same three tools as always: randomization, bootstrapping, and the t-distribution mathematical model.

Case study: two exam versions

An instructor gives two versions of an exam, shuffled randomly to students. Version B students worry theirs was harder.

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$H_0 : \mu_A - \mu_B = 0$ (equally difficult) vs. $H_A : \mu_A - \mu_B \neq 0$. Observed difference: 3.1 points.

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Randomizing exam labels 1,000 times: differences due to chance alone ranged from about -10 to $+10$ points — so 3.1 doesn't stand out much.

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This does **not** mean the exams are equally difficult — it means the data can't distinguish between “equally difficult” and “Exam A is a bit easier.” Hypothesis testing is built to reject a claim, not to prove one. In this case, the teacher (and class) can be glad there's no evidence one exam was harder.

Bootstrapping the difference: stem cells and heart function

Sheep with heart attacks, randomized to embryonic stem cell (ESC) treatment or control; outcome is change in heart pumping capacity.

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Bootstrap each group separately (with replacement), subtract the means, repeat thousands of times — builds a confidence interval with **no null hypothesis assumed**.

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Since this was a randomized experiment, we conclude ESC treatment **causes** improved heart pumping capacity relative to control.

The mathematical model: t-test for two means

i T statistic for two independent means

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad df = \min(n_1 - 1, n_2 - 1)$$

Conditions: independent observations within and between groups; large samples with no extreme outliers (checked separately per group).

Worked example: smoking and birth weight

Does maternal smoking relate to newborn birth weight? ($n = 1000$ births sampled)

Group	n	Mean	SD
Nonsmoker	867	7.27	1.23
Smoker	114	6.68	1.60

$$\bar{x}_n - \bar{x}_s = 0.59, SE = \sqrt{1.23^2/867 + 1.60^2/114} = 0.16.$$

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Two-sided p-value ≈ 0.00034 . Since $p < 0.05$, **reject** H_0 — convincing evidence of a real difference in average birth weight.

Comparing paired means

When independence fails on purpose

Sometimes two measurements are **linked** by design: the same subject measured twice, or naturally matched pairs (same car, same textbook, same day).

i Paired data

Two sets of observations are paired if each observation in one set corresponds to exactly one observation in the other.

The key move: collapse each pair into a single number — its **difference** — and suddenly this is just a **one-sample** problem (Chapter 19) again.

Case study: tire tread wear

25 cars, each fitted with one Smooth Turn tire and one Quick Spin tire. Mean tread: Quick Spin 0.308 cm, Smooth Turn 0.310 cm — a tiny raw difference.

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The observed difference falls **far outside** the randomization distribution (which ranges roughly ± 0.0025). We reject H_0 — the brands really do differ in average tread wear.

Bootstrapping paired differences: textbook prices

201 UCLA courses sampled; 68 books found on both the UCLA Bookstore and Amazon.
Treat each book's **price difference** (Bookstore — Amazon) as a single data point.

n	Mean	SD
68	\$3.58	\$13.4

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68	\$3.58	\$13.4

Bootstrapping the 68 differences (with replacement) gives a 99% percentile CI of about (\$0.25, \$7.87) — suggesting the Bookstore tends to run higher, though the exact margins are uncertain.

The t-test on differences

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$$T = \frac{\bar{x}_{diff} - 0}{s_{diff}/\sqrt{n_{diff}}}, \quad df = n_{diff} - 1$$

For the textbook data: $SE = 13.42/\sqrt{68} = 1.63$, $T = 3.58/1.63 = 2.20$, $df = 67$.

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Two-sided p-value $\approx 0.031 < 0.05$ — reject H_0 . 95% CI:

$3.58 \pm 2.00(1.63) = (\$0.32, \$6.84)$ — the UCLA Bookstore is, on average, higher-priced than Amazon.

A note on power

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But pairing isn't always available — you need a real, natural correspondence between observations, not just equal sample sizes.

Comparing many means: ANOVA

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If you run enough pairwise comparisons, **you will eventually find “significance” somewhere just by chance**, even when nothing real is going on — this is **data snooping** / the **multiple comparisons problem**.

i ANOVA (ANalysis Of VAriance)

A single, holistic test of whether *at least one* group mean differs — before you go looking for which one.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \qquad H_A : \text{at least one } \mu_j \text{ differs}$$

The core idea

ANOVA compares two kinds of variability:

- ▶ **MSG** (mean square between groups) — how much the group *means* differ from the overall mean.
- ▶ **MSE** (mean square error) — how much individual observations vary *within* each group.

$$F = \frac{MSG}{MSE}$$

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$$F = \frac{MSG}{MSE}$$

If H_0 is true, both should be similar in size, so $F \approx 1$. A big F means the between-group differences are large *relative to* the natural noise within groups.

Case study: baseball batting by position

Do on-base percentages (OBP) differ by fielding position?

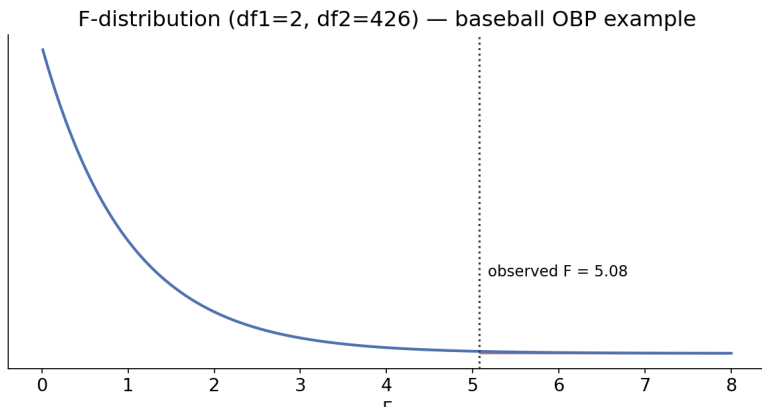
Position	n	Mean	SD
OF	160	0.320	0.043
IF	205	0.318	0.038
C	64	0.302	0.038

$MSG = 0.00803$, $MSE = 0.00158$, so $F = MSG/MSE \approx 5.08$.

The F-distribution

i F-test

F follows an F -distribution with $df_1 = k - 1$ (groups) and $df_2 = n - k$ (residual) when H_0 is true.



P-value

Here: $df_1 = 3 - 1 = 2$, $df_2 = 429 - 3 = 426$.

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p-value $\approx 0.0066 < 0.05$ — **reject** H_0 : average OBP genuinely varies by position.
(ANOVA doesn't say *which* position differs — that requires careful follow-up, avoiding the multiple-comparisons trap.)

Reading software output

term	df	sumsq	meansq	statistic	p.value
position	2	0.0161	0.0080	5.08	0.0066
Residuals	426	0.6740	0.0016		

In R: `aov(OBP ~ position, data = mlb_players_18) %>% summary()` (or `anova(lm(...))`) produces exactly this table.

Conditions for ANOVA

- ▶ **Independence** — within and across all groups.
- ▶ **Approximately normal** within each group (or large enough samples with no extreme outliers).
- ▶ **Constant variance** across groups — check with side-by-side boxplots; a red flag if one group's spread looks much larger than another's.

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Notice ANOVA gives **no confidence interval** — there's no single “difference” parameter when comparing 3+ groups at once. For that, drop back down to Chapter 20's two-mean methods, applied carefully (not comparing every possible pair).

Extending the exam example to three exams

Recall Exam A vs. B (Ch. 20) — now imagine a third version, Exam C:

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$H_0 : \mu_A = \mu_B = \mu_C$. Randomizing 1,000 times gives an observed $F = 3.48$, with simulated p-value ≈ 0.036 .

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Reject H_0 — some exam is genuinely different in difficulty. But careful: this doesn't mean Exam C specifically is harder — concluding that from eyeballing the means, after the fact, is exactly the data-snooping trap ANOVA exists to avoid.

Wrapping up

- ▶ **Two independent means:** same $\bar{x}_1 - \bar{x}_2$ point estimate, same t-distribution logic as one mean — just a new SE formula combining both groups' variability.
- ▶ **Paired means:** the pairing structure lets you collapse two dependent measurements into one difference — turning a two-sample problem back into a one-sample problem, often with more power.
- ▶ **Many means (ANOVA):** a single test to check “is *any* group different?” before you go hunting for which one — protecting against inflated Type I error from multiple comparisons.
- ▶ All three still rest on the same foundation from Ch. 11-14: a point estimate, a measure of its natural variability, and a way to ask “could chance alone explain what we're seeing?”